

Seeing Small Signs: Does Super-Resolution Help Traffic Sign Detection?

An Empirical Study of SR Preprocessing Across CNN and Transformer Detectors

Team 23

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1. Task and Motivation

Traffic sign detection is a safety-critical component of autonomous driving and advanced driver-assistance systems (ADAS). Despite significant progress in object detection, performance degrades substantially on low-quality imagery — images captured under adverse conditions such as motion blur, rain, low light, or at long range. In these settings, signs appear as small, degraded patches in the scene, making them hard to localize and classify reliably.

Super-resolution (SR) preprocessing offers a principled approach to recovering high-frequency spatial detail prior to detection. The central question this project asks is: *does applying SR before detection measurably improve traffic sign detection accuracy, and if so, which SR model and detector combination benefits most?*

This question is practically motivated. Autonomous systems must process video in real time, meaning any SR step must not compromise throughput. We therefore evaluate not only accuracy gains but also the end-to-end latency introduced by SR preprocessing, establishing whether a given pipeline remains viable under a 25 FPS real-time threshold.

A secondary motivation is architectural. Pixel-space SR models (ESPCN [8], FSRCNN [2]) preprocess the full image before the detector sees it. AnyUp [7], by contrast, is a universal feature-level upsampler that replaces the upsampling operator inside the detector backbone, operating on intermediate feature maps rather than raw pixels. Unlike pixel-space SR, AnyUp integrates at the feature level between the backbone and the detection head, and is applicable to all three detector architectures studied. Comparing these two integration paradigms across CNN and transformer detectors yields insight into where SR provides the greatest benefit.

2. Goals

The project has three concrete research goals:

G1 – Accuracy. Determine whether $2\times$ SR preprocessing improves traffic sign detection accuracy (mAP@0.5)

relative to a no-SR baseline, across all four detector configurations.

G2 – Scale sensitivity. Determine whether accuracy gains concentrate on small-scale signs, which is the primary hypothesis motivating SR-as-preprocessing. We decompose mAP by COCO object scale bins (small / medium / large).

G3 – Real-time viability. Measure end-to-end pipeline FPS (SR + detection) for every condition and identify which configurations meet the 25 FPS real-time threshold.

3. Methods

3.1. Pipeline Overview

The pipeline has two stages. In Stage 1, each input image is passed through a SR model at $2\times$ scale to produce an upsampled version. In Stage 2, each detector runs on the SR-preprocessed image (or the raw baseline). This defines a full 4×4 pipeline matrix with 16 conditions, shown in Tab. 1.

SR models:

- *None (baseline)* — raw degraded image, no preprocessing.
- *ESPCN $2\times$* [8] — efficient sub-pixel convolutional upsampler; operates on raw pixels.
- *FSRCNN $2\times$* [2] — compact CNN SR model; operates on raw pixels.
- *AnyUp* [7] — universal feature-level upsampler; replaces the upsampling operator inside each detector backbone. Integration points: between the CSP backbone and PAN neck in YOLOv8; between ResNet-50 and FPN in Faster R-CNN; between the ResNet backbone and transformer encoder in DETR.

Detectors:

- *YOLOv8n* [4] — fine-tuned on traffic sign classes via Ultralytics.

- *Faster R-CNN* [6] — pretrained ResNet-50 backbone from torchvision; RPN and detection head fine-tuned.
- *DETR* (pretrained) [1] — facebook/detr-resnet-50 from HuggingFace; evaluated zero-shot or with minimal adaptation.
- *DETR* (fine-tuned) — same architecture, fully fine-tuned on the traffic sign training split.

SR Model	YOLOv8n	F-RCNN	DETR-PT	DETR-FT
None (baseline)	✓	✓	✓	✓
ESPCN 2×	✓	✓	✓	✓
FSRCNN 2×	✓	✓	✓	✓
AnyUp	✓	✓	✓	✓

Table 1. Full 4×4 pipeline matrix (16 conditions $\times$ 2 datasets = 32 runs). PT = pretrained, FT = fine-tuned. AnyUp integrates at the feature level inside each backbone and is applicable to all detectors.

3.2. AnyUp Integration

AnyUp [7] is a learned, content-adaptive upsampling module that can replace any fixed upsampling operator (bilinear, nearest-neighbour) in a network. For each detector we identify the natural upsampling hook point and substitute AnyUp there, keeping all other weights frozen during a short adaptation phase:

YOLOv8n: AnyUp replaces the upsample layer at the input of the PAN neck, upsampling the coarsest CSP backbone feature map before multi-scale fusion.

Faster R-CNN: AnyUp replaces the upsample step between the ResNet-50 backbone and the FPN, prior to region proposal generation.

DETR: AnyUp is inserted between the ResNet-50 backbone output and the transformer encoder input, providing the clearest architectural separation.

3.3. Training Setup

All experiments run on GPU. YOLOv8n is trained with the Ultralytics API. Faster R-CNN and DETR use PyTorch and HuggingFace Transformers respectively. ESPCN and FSRCNN use pretrained weights without further SR fine-tuning. All detectors are trained and evaluated on a consistent 80/20 stratified split per dataset.

4. Datasets

We use two publicly available traffic sign benchmarks:

GTSDDB [3] — The German Traffic Sign Detection Benchmark. Contains 900 full-scene images with 1213 annotated signs across 43 categories, at varying scales and real-world capture conditions.

TT100K [9] — The Tsinghua-Tencent 100K dataset. Contains 100,000 Tencent Street View images with over 30,000 traffic sign instances across 232 categories, including many small and partially occluded signs under diverse lighting and weather conditions.

Both datasets consist of real captured images with no paired high-resolution ground truth. This rules out pixel-level SR quality metrics (PSNR, SSIM, LPIPS), which require an HR reference image. All evaluation is therefore performed at the detection level only.

5. Evaluation

5.1. Metrics

Primary metric: mAP@0.5 — mean average precision at IoU threshold 0.5, reported per condition and per dataset.

Secondary metrics:

- *mAP by object scale* — mAP@0.5 decomposed into small ($< 32^2$ px), medium (32^2 – 96^2 px), and large ($> 96^2$ px) bins following the COCO convention [5]. This is the most diagnostic breakdown: if SR improves overall mAP without improving small-object mAP, the core hypothesis does not hold.
- *End-to-end FPS* — wall-clock throughput from raw input to predicted boxes, SR included. Hard threshold of 25 FPS for real-time viability, reported per pipeline combination.

5.2. Research Questions

The 16-condition matrix is designed to answer three research questions:

1. **Does SR help detection at all?** Baseline vs. all SR conditions on overall mAP.
2. **Which SR model helps most?** ESPCN vs. FSRCNN (pixel-space) vs. AnyUp (feature-level) on mAP and FPS across all detectors.
3. **Which detector benefits most?** CNN-based (YOLOv8n, Faster R-CNN) vs. transformer-based (DETR pretrained and fine-tuned) under equivalent SR conditions.

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